Chapter

6

Industrial Microbiology

Learning Objectives

After studying this chapter the students will be able to,

- *Know the concepts involved in industrial microbiology and the production of industrially microbiology production by microorganisms.*
- *Understand the primary and secondary screening process.*
- *Gain knowledge in the field of strain improvement of microorganisms.*
- *Describe the structure components and function of a fermentor.*
- *Known the principles behind fermentation medium, fermentation process, upstream processing and downstream processing.*
- *Know the values of microorganism used in the production of penicillin citric acid, wine and single cell protein.*
- *Analyze the basics behind immobilization of microorganisms*

Chapter Outline

6.1 Industrially Important Micro Organisms and their Products

6.2 Screening of Industrially Important Microorganism

6.3 Strain Improvement

6.4 Preservation of Industrially Important Microorganisms

6.5 Fermentors

6.6 Industrial Production of Penicillin

6.7 Industrial Production of Wine

6.8 Industrial Production of Single Cell Protein

6.9 Industrial Production of Citric Acid

6.10 Immobilization

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Industrial microbiology is a branch of science that deals with the study and uses of various microorganisms that are responsible for the production of many products which has industrial and economic applications. Man has been using many microorganisms for the production of foods, (bread, cheese, yoghurt, pickles)–beverages (beer, wine) for many centuries. The birth of industrial microbiology largely began with the studies of Pasteur on fermentation. The term Fermentation originates from a Latin verb "Fervere" which literally means to boil. In alcohol production, CO_2 (gas bubbles) Figure 6.1 are formed during boiling of liquid.

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